

– TECHNICAL DATA SHEET

NN-RockGrout Resin Capsule

Resin Capsule — Grout Material for Roof Bolting in Mines

Pre-packed two-component polyester resin capsules, supplied in protective cartons



Pre-packed polyester resin grout capsules for mine roof bolting

Document No.	Revision	Issue Date	Product Code
NNPL/TDS/PROD-04	1.0	29 June 2026	RC-04

01 PRODUCT DESCRIPTION

NN-RockGrout Resin Capsule is a self-contained two-component polyester resin cartridge used as a grouting medium for roof bolting and strata control in underground coal mines and tunnels. Each capsule holds a precisely proportioned unsaturated-polyester resin mastic (mineral-filled) alongside a separately sealed organic-peroxide catalyst, contained in a sausage-shaped film pouch.

In use, the capsule is inserted into the drilled borehole and the roof bolt is spun through it, rupturing the partition film and shearing the resin and catalyst together. This triggers a rapid exothermic cure that anchors the bolt to the strata. The product is designed to comply with the testing and performance requirements of DGMS Circular (Approval) No. 04 of 2013 for resin capsules used as grouting material for roof bolting in mines.

02 KEY FEATURES & BENEFITS

- **DGMS-compliant** — Formulated to meet DGMS Circular (Approval) No. 04 of 2013 for resin-capsule grout in belowground coal mines.
- **Fast & slow set** — Fast and slow-set grades give controlled gel/set times for staged anchoring in the bolting cycle.

- **High anchorage** — Develops high anchorage and compressive strength for reliable roof support.
- **No site mixing** — No on-site batching — pre-packed and pre-proportioned; the spinning bolt performs the mix.
- **Durable & low-flammability** — Cured resin shows very low flammability and resists groundwater, protecting the anchored bolt.
- **Clearly marked** — Each capsule is superscripted with set type and gel/setting time for correct selection.

03 REACTION & SETTING CHARACTERISTICS

Per DGMS Circular (Approval) No. 04 of 2013, Para 2.0 (A).

Parameter	Standard
Gel & setting time — Fast-set type	23 seconds (min) – 54 seconds (max)
Gel & setting time — Slow-set type	134 seconds (min) – 202 seconds (max)
Reaction temperature	Not more than 80 °C
Thermal stability (5 °C / 1 hr and 45 °C / 1 hr)	Gel & set times remain within specified limits

04 PHYSICO-MECHANICAL PROPERTIES

Per DGMS Circular (Approval) No. 04 of 2013, Para 2.0 (B). Applicable for slow-set type resin.

Parameter	Standard
Compressive strength — after 30 minutes	≥ 30.0 MPa
Compressive strength — after 24 hours	≥ 80.0 MPa
Bond strength — after 30 minutes	≥ 10.0 tonnes
Bond strength — after 24 hours	≥ 15.0 tonnes
Shear strength of set resin	≥ 19.0 MPa
Shrinkage — after 24 hours	Maximum 0.01 %
Shrinkage — after 7 days	Maximum 0.01 %

05 OTHER PROPERTIES

Per DGMS Circular (Approval) No. 04 of 2013, Para 2.0 (C).

Parameter	Standard
Flammability	Very low flammability (passes flame test)
Reaction temperature ceiling	≤ 80 °C
Standard capsule reference size	24 mm dia × 600 mm (or equivalent)

06 SHORT ENCAPSULATION PULL TEST (SEPT)

User-end verification per DGMS Circular (Approval) No. 04 of 2013, Annexure-3.

Condition	Minimum performance
After 30 minutes — 20 mm dia roof bolt	≥ 10.0 tonnes
After 30 minutes — 22 mm dia roof bolt	≥ 12.0 tonnes
After 8 or 24 hours	≥ 15.0 / 20.0 tonnes respectively
Net displacement measured during SEPT	Not more than 15 mm

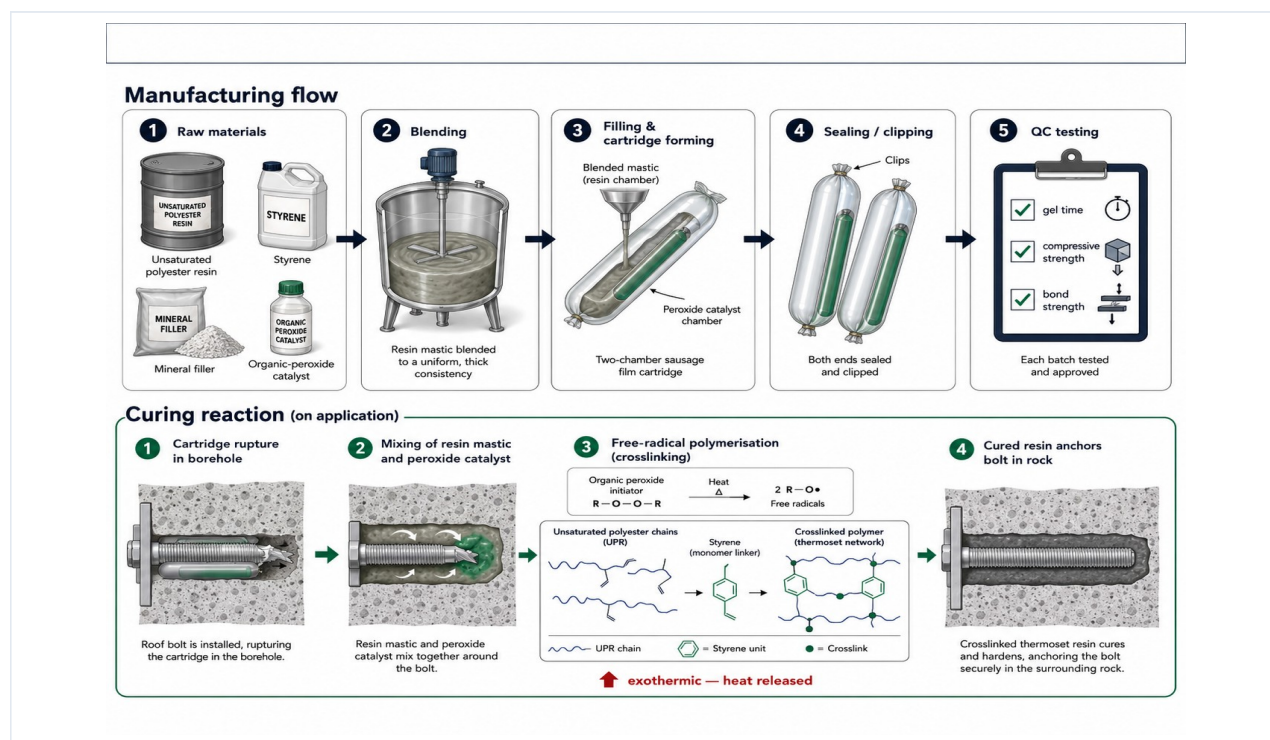
07 APPLICATION & INSTALLATION



Resin-grouted roof-bolt installation into the strata in an underground coal mine

- **Drill** — Drill the borehole to the specified depth and diameter; ensure it is clean and the annulus suits the capsule for proper mixing.
- **Insert** — Insert the required number of capsules to the back of the hole; for mixed systems, load fast-set toward the toe.
- **Spin & mix** — Push the bolt to puncture the capsules, then spin it to fully shear and mix the resin and catalyst for the specified spin time.
- **Hold** — Hold the bolt static during gel; do not disturb until the set time has elapsed.
- **Tension** — Apply tension/torque only after the resin has reached the required anchorage strength.

08 MANUFACTURING & CURE PROCESS



Manufacturing flow and on-application cure (polyester resin + peroxide catalyst → crosslinked thermoset)

During cure, the organic peroxide initiates free-radical polymerisation of the styrene and unsaturated-polyester chains, crosslinking the mineral-filled mastic into a hard, high-strength thermoset that bonds the bolt to the rock.

09 PACKAGING, STORAGE & SHELF LIFE

Item	Detail
Capsule sizes	Made to order in diameters and lengths to suit the bolt / borehole
Packaging	Individually formed cartridges in cartons; each capsule superscripted with set type and gel/set time
Storage	Store cool, dry and shaded, away from heat, sparks and direct sunlight; rotate stock (FIFO)
Handling	Protect every consignment from extreme weather that could alter resin properties
Re-testing	Re-test gel/set time of the oldest stored batch at intervals not exceeding one month

10 COMPLIANCE & TESTING

Each batch is manufactured and tested in line with DGMS Circular (Approval) No. 04 of 2013. Gel & setting time, reaction temperature, thermal stability, compressive, bond and shear strength, shrinkage and flammability are tested per the methods in Annexure-1 of the circular. One lot is sampled per 10,000 capsules; batch records, usage and storage manuals are maintained and supplied with consignments as required.

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